WEEK 2 INTERNSHIP REPORT

Supervised Machine Learning — Student Marks Prediction

Internship	YuvaIntern — AI Pioneers Internship
Task	Week 2 — Supervised Machine Learning Models
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GitHub repository	https://github.com/253115243036-jpg/ai-pioneers-week2-preprocessing

1. Abstract

This project demonstrates a beginner-friendly supervised machine-learning workflow that predicts student examination marks from study hours. A small illustrative dataset is created with Python, then Linear Regression and Decision Tree Regression models are trained and compared using Mean Absolute Error (MAE) and R^2 score. The comparison is exported to a CSV file and predictions are visualized.

2. Objectives

• Create and inspect a small dataset containing study hours and marks.

• Separate input features and target values.

• Split the data into training and testing sets.

• Train Linear Regression and Decision Tree Regression models.

• Compare model performance using MAE and R^2 score.

• Save the comparison results and a visualization.

3. Dataset description

The dataset is manually created for demonstration and contains 10 observations. The input feature is Study_Hours and the target variable is Marks. This is sample data created for learning; it is not real student data and should not be used to make claims about actual students.

4. Tools and libraries

Python, Google Colab, Pandas, Matplotlib, and scikit-learn.

5. Methodology

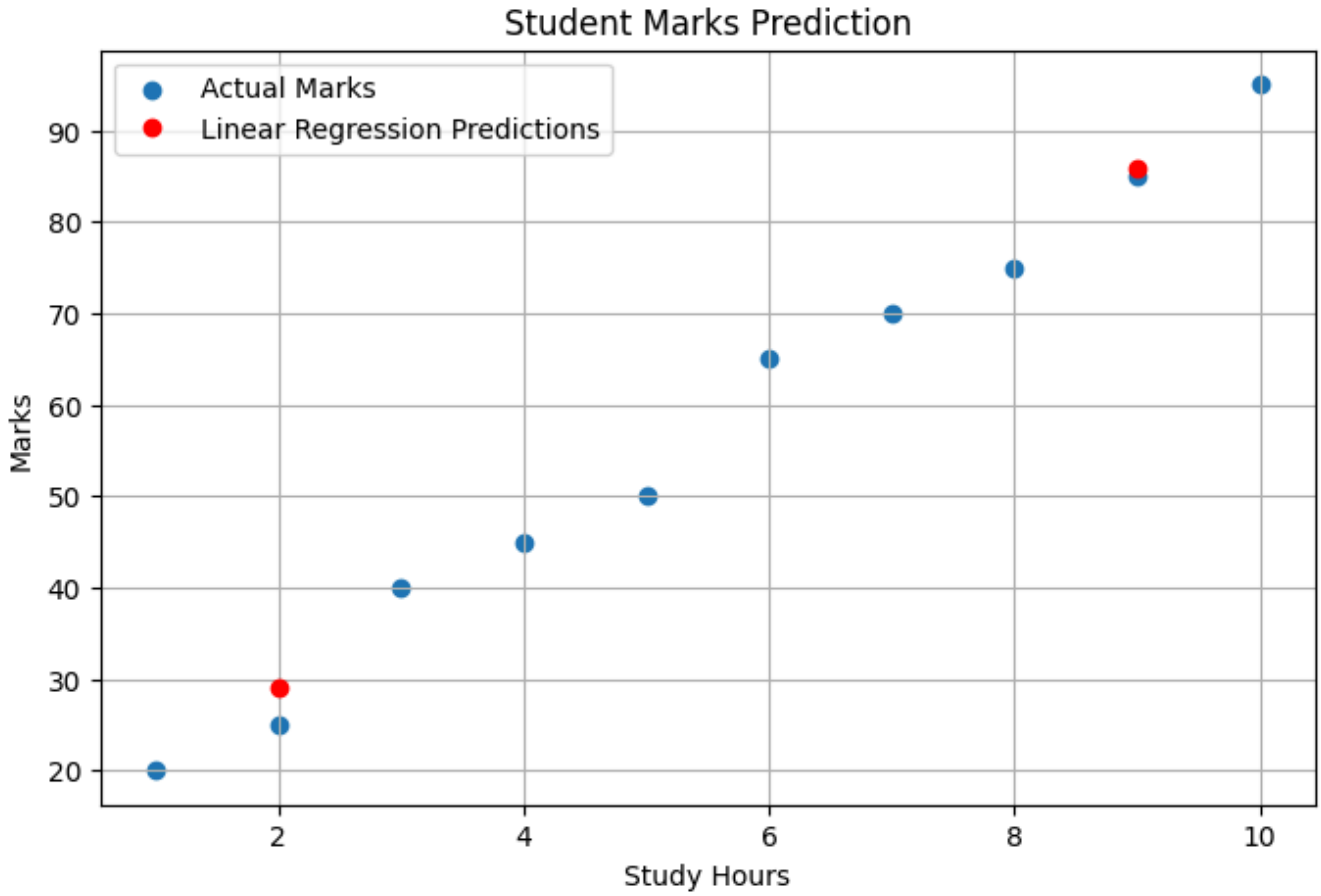
Data creation: a Pandas DataFrame is created from the sample study-hours and marks values. Feature/target selection: Study_Hours is used as X and Marks as y. Train-test split: the data is split using test_size=0.2 and random_state=42. Model training: LinearRegression and DecisionTreeRegressor(max_depth=3, random_state=42) are trained on the same training set. Evaluation: both models are compared using MAE and R^2 score. Outputs: Model_Comparison.csv and Student_Marks_Graph.png are saved.

6. Results

Complete the table below using the actual output from your Colab notebook. Do not estimate or invent values.

Measure	Actual result from Colab
Linear Regression — MAE	2.50
Linear Regression — R^2 score	0.99
Decision Tree Regression — MAE	8.75
Decision Tree Regression — R^2 score	0.90
Best model on this test split	Linear Regression may be more suitable for this project because it is simple, easy to interpret, and works well when study hours and marks have a roughly linear relationship. However, the final choice should be based on comparing both models' MAE and R^2 scores on the test data.

7. Visualization and observations



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Observation:The graph shows a positive relationship between study hours and students' marks. As study hours increase, marks generally increase. The Linear Regression model attempts to represent this trend with a straight line, while the Decision Tree Regression model predicts marks using decision-based patterns. This suggests that study hours may influence academic performance. However, the dataset is small and illustrative, so further testing with real data is needed.

8. Output files

The notebook is expected to produce the following files after the relevant cells run successfully:

- Model_Comparison.csv — comparison of both regression models.
- Student_Marks_Graph.png — plot of the sample data and Linear Regression predictions.
- Week2_Student_Marks_Prediction.ipynb — downloaded Colab notebook.
- README.md — project overview and instructions.
- Week2_Report.pdf — completed report.

9. Conclusion

This task provides practical experience with supervised regression, training/testing data splits, model prediction, evaluation metrics, and visualization. The two models were evaluated on the same test split to support a fair comparison. Since the dataset is very small and illustrative, the results are only a learning demonstration and cannot be generalized to real student performance.

10. References

- scikit-learn Linear Regression: https://scikit-learn.org/stable/modules/generated/sklearn.linear_model.LinearRegression.html
- scikit-learn Decision Tree Regression: https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeRegressor.html
- scikit-learn model evaluation: https://scikit-learn.org/stable/modules/model_evaluation.html
- Pandas documentation: https://pandas.pydata.org/docs/
- Google Colab: https://colab.research.google.com/